Physics 5A Midterm 1 Review Session

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Use $g = 9.8\,\mathrm{m/s^2}$ and neglect air resistance in problems 1 and 2. Make sure you show your work and include units in your final answers.

Problem 1: Jinyan throws an eraser directly upward with an initial speed $5.00\,\mathrm{m/s}$. The eraser travels up and then comes back down. Jinyan catches it on its way down, and its speed just before he catches it is $6.00\,\mathrm{m/s}$. Take the positive y-direction to be upward.

Part (a): Find the vertical displacement of the eraser from the moment it leaves Jinyan's hand to the moment just before he catches it. Does he catch it above or below its release point?

Part (b): Find the maximal height the eraser can reach, measured with respect to its release point.

Part (c): Find the total time the eraser spends in the air before Jinyan catches it.

Part (d): Find the total distance that the eraser has traveled in the air.

Problem 2: During the discussion this week, Jinyan threw an eraser into the air at an angle $45°$ above the horizontal, toward the right. The eraser hit the ground 1.00 second after it left his hand. Its vertical displacement from the release point to the ground was $-1.40\,\mathrm{m}$. Take the positive x-direction to be to the right, and the positive y-direction to be upward.

Part (a): Find the y-component of the eraser's initial velocity.

Part (b): Find the x-component of the eraser's initial velocity.

Part (c): Find the x-component of the eraser's velocity when it reaches the maximal height of its trajectory.

Problem 3: A rocket in space has two thrusters. The first thruster provides an acceleration $\vec{a}_1$ with magnitude $6.00\,\mathrm{m/s^2}$ at an angle 30° above the positive x-direction. The second thruster provides an acceleration $\vec{a}_2$ with magnitude $2\sqrt{3}\,\mathrm{m/s^2}$ at an angle 60° below the positive x-direction. The total acceleration of the rocket is the vector sum of these two accelerations. Neglect gravity and any other forces in this problem.

Part (a): Find the x- and y-components of $\vec{a}_1$ and $\vec{a}_2$. Then find the total acceleration $\vec{a} = \vec{a}_1 + \vec{a}_2$, written in terms of $\hat{x}$ and $\hat{y}$.

Part (b): Find the magnitude and direction of the total acceleration $\vec{a}$. Write its direction as an angle measured from the positive x-direction.

Part (c): Now the magnitude of $\vec{a}_2$ is reduced by half, while its direction and the entire vector $\vec{a}_1$ stay the same. Find the components of the new acceleration from the second thruster, then compute the new total acceleration vector $\vec{a}\,'$.

Part (d): Find the magnitude and direction of $\vec{a}\,'$. Does the total acceleration still point in the positive x-direction? Explain your answer.

Solutions

Problem 1

I recommend first writing down the initial velocity, final velocity, and acceleration with their correct signs. The eraser initially moves upward, but it is moving downward just before Jinyan catches it. Therefore,

$$v_{0,y} = +5.00\,\mathrm{m/s}\,, \qquad v_{\mathrm{f},y} = -6.00\,\mathrm{m/s}\,, \qquad a_y = -9.8\,\mathrm{m/s^2}\,.$$

Notice that the problem gives us speeds, which are positive numbers. We need to include the direction when writing the corresponding velocities. Here we set the release point to be $y_0 = 0\,\mathrm{m}$.

Part (a): We know the initial velocity, final velocity, and constant acceleration, but we do not yet know the time. Thus we use the kinematic equation

$$v_{\mathrm{f},y}^2 = v_{0,y}^2 + 2a_y(y_\mathrm{f} - y_0)\,. \tag{‡}$$

Writing $\Delta y = y_\mathrm{f} - y_0$ and isolating it, we obtain

$$\Delta y = \frac{v_{\mathrm{f},y}^2 - v_{0,y}^2}{2a_y} = \frac{(-6.00\,\mathrm{m/s})^2 - (5.00\,\mathrm{m/s})^2}{2(-9.8\,\mathrm{m/s^2})} = \frac{36.0 - 25.0}{-19.6}\,\mathrm{m} = -0.561\,\mathrm{m}\,.$$

The negative sign means that Jinyan catches the eraser 0.561 meters below its release point. This is the displacement between the two endpoints; it does not count the whole distance traveled up and down.

However, as I have mentioned during discussion, the following two kinematic equations have captured the full physics,

$$\Delta y = \frac{1}{2}a_y(\Delta t)^2 + v_{0,y}\,\Delta t\,, \qquad \Delta v_y = a_y\,\Delta t\,, \tag{†}$$

so we could have solved this part using these two equations only. If we were doing it in this way, we would first need a time label t_f for Jinyan catching the eraser. Let us set the time the eraser leaves Jinyan's hand as $t_0 = 0$; then

$$v_{\mathrm{f},y} - v_{0,y} = \Delta v_y = a_y\,\Delta t = a_y(t_\mathrm{f} - t_0)\,,$$

from which we can plug in numbers and get

$$t_\mathrm{f} - t_0 = \frac{v_{\mathrm{f},y} - v_{0,y}}{a_y} = \frac{-6.00\,\mathrm{m/s} - 5.00\,\mathrm{m/s}}{-9.8\,\mathrm{m/s^2}} = \frac{11.0}{9.8}\,\mathrm{s} \approx 1.12\,\mathrm{s}\,.$$

Since $t_0 = 0$, this gives $t_\mathrm{f} = 1.12\,\mathrm{s}$. Then we can calculate the vertical displacement,

$$\begin{aligned}\Delta y &= \frac{1}{2}a_y(t_\mathrm{f} - t_0)^2 + v_{0,y}(t_\mathrm{f} - t_0) \\ &= \frac{1}{2}(-9.8\,\mathrm{m/s^2})\left(\frac{11.0}{9.8}\,\mathrm{s}\right)^2 + (5.00\,\mathrm{m/s})\left(\frac{11.0}{9.8}\,\mathrm{s}\right) = -0.561\,\mathrm{m}\,.\end{aligned}$$

This agrees with the result above. We first used the change in velocity to find the elapsed time, then used that time to find the displacement.

Problem 1 (continued)

Part (b): When the eraser reaches its maximal height, its vertical velocity becomes zero because it is about to turn around. We can again solve this part using only the two equations (†). The ending point is now the top of the trajectory, not the point where Jinyan catches it. Let t_{top} be the time when it reaches the top. First we use the second equation in (†),

$$0 - v_{0,y} = a_y(t_{\text{top}} - t_0)\,.$$

Plugging in the numbers, we see

$$t_{\text{top}} = \frac{0 - 5.00\,\text{m/s}}{-9.8\,\text{m/s}^2} = \frac{5.00}{9.8}\,\text{s} \approx 0.510\,\text{s}\,.$$

Now we use the first equation in (†) for this upward portion of the journey. Denoting the height above the release point as $h = y_{\max} - y_0$, we find

$$\begin{aligned} h &= \frac{1}{2}a_y(t_{\text{top}} - t_0)^2 + v_{0,y}(t_{\text{top}} - t_0) \\ &= \frac{1}{2}(-9.8\,\text{m/s}^2)\left(\frac{5.00}{9.8}\,\text{s}\right)^2 + (5.00\,\text{m/s})\left(\frac{5.00}{9.8}\,\text{s}\right) \approx 1.28\,\text{m}\,. \end{aligned}$$

Thus the eraser reaches 1.28 meters above its release point.

Part (c): This is the elapsed time $t_{\text{f}} - t_0$ that we already found in part (a). Using the kinematic equation that relates the change in velocity to the elapsed time,

$$v_{\text{f},y} - v_{0,y} = a_y\,\Delta t\,,$$

we see that

$$\Delta t = \frac{v_{\text{f},y} - v_{0,y}}{a_y} = \frac{-6.00\,\text{m/s} - 5.00\,\text{m/s}}{-9.8\,\text{m/s}^2} = 1.12\,\text{s}\,.$$

This is the total time from release to catching, including both the upward and downward portions.

Part (d): To find the total distance, we need to add the distance traveled upward and the distance traveled downward. The upward distance is h from part (b). The eraser then falls from the top to the catching point, which is below the release point, so the downward distance is $y_{\max} - y_{\text{f}} = h - \Delta y$. That is,

$$d_{\text{up}} = h = \frac{25.0}{19.6}\,\text{m}\,, \qquad d_{\text{down}} = h - \Delta y = \left(\frac{25.0}{19.6} - \left(-\frac{11.0}{19.6}\right)\right)\,\text{m} = \frac{36.0}{19.6}\,\text{m}\,.$$

Therefore, the total distance traveled is

$$d_{\text{total}} = d_{\text{up}} + d_{\text{down}} = \frac{25.0 + 36.0}{19.6}\,\text{m} = 3.11\,\text{m}\,.$$

Both distances are positive, even though the eraser moves in opposite directions during the two portions. The total distance is 3.11 m, while the vertical displacement is -0.561 m.

Problem 2

Once the eraser leaves Jinyan's hand, gravity is the only thing trying to change its motion. Gravity points downwards, so the acceleration vector is

$$\vec{a} = (a_x,\, a_y) = (0,\, -9.8)\,\mathrm{m/s}^2\,.$$

There is nothing pulling or pushing the eraser in the horizontal direction. We should therefore write separate kinematic equations for the two directions. With initial time t_0 set to be zero at the moment the eraser leaves Jinyan's hand, we can write

$$\Delta x = v_{0,x}t + \frac{1}{2}a_x t^2 = v_{0,x}t\,,$$
$$\Delta y = v_{0,y}t + \frac{1}{2}a_y t^2\,.$$

Never mix the x- and y-components; treat the two motions independently. The time t is the same in both equations because the horizontal and vertical motions happen at the same time.

Part (a): The given displacement $\Delta y = -1.40\,\mathrm{m}$ is a vertical displacement, so we use only the equation in the y-direction. Isolating $v_{0,y}$ gives

$$v_{0,y} = \frac{\Delta y - a_y t^2/2}{t} = \frac{-1.40\,\mathrm{m} - (-9.8\,\mathrm{m/s}^2)(1.00\,\mathrm{s})^2/2}{1.00\,\mathrm{s}} = 3.50\,\mathrm{m/s}\,.$$

The positive sign tells us that the eraser was initially moving upward. Notice that we did not need its horizontal velocity or its launch angle to do this calculation.

Part (b): The initial velocity vector makes an angle $\theta = 45^\circ$ above the positive x-direction. Using the right triangle formed by the velocity vector and its components, we have

$$v_{0,x} = v_0\cos(\theta)\,, \qquad v_{0,y} = v_0\sin(\theta)\,.$$

We already know $v_{0,y} = 3.50\,\mathrm{m/s}$ from part (a), so we first use the second equation to find the initial speed v_0,

$$v_0 = \frac{v_{0,y}}{\sin(\theta)} = \frac{3.50\,\mathrm{m/s}}{\sin(45^\circ)} = 3.50\sqrt{2}\,\mathrm{m/s} \approx 4.95\,\mathrm{m/s}\,.$$

Then we use the first equation to find the horizontal component. Keeping the exact expression for v_0,

$$v_{0,x} = v_0\cos(\theta) = (3.50\sqrt{2}\,\mathrm{m/s})\cos(45^\circ) = (3.50\sqrt{2}\,\mathrm{m/s})\frac{1}{\sqrt{2}} = 3.50\,\mathrm{m/s}\,.$$

Thus the initial velocity vector is $\vec{v}_0 = (3.50,\, 3.50)\,\mathrm{m/s}$. Here, gravity changes only the vertical component during the flight.

Part (c): At the maximal height, $v_y = 0$, but this does not tell us that $v_x = 0$. We need to use the horizontal acceleration to find the horizontal velocity,

$$v_x = v_{0,x} + a_x t = v_{0,x} + 0 = 3.50\,\mathrm{m/s}\,.$$

This holds throughout the flight, including at the top. Therefore, the velocity vector at the maximal height is $\vec{v} = (3.50,\, 0)\,\mathrm{m/s}$. The eraser is <u>not at rest</u>: it is still moving to the right at $3.50\,\mathrm{m/s}$, even though it has momentarily stopped moving up or down.

Problem 3

Part (a): I recommend drawing the two acceleration vectors first. The first points up and to the right, so both of its components are positive. The second points down and to the right, so its x-component is positive and its y-component is negative.

Since the angles are measured from the positive x-direction, we use cosine to find the x-components and sine to find the y-components. For the first thruster,

$$a_{1,x} = (6.00\,\mathrm{m/s^2})\cos(30^\circ) = 6.00 \cdot \frac{\sqrt{3}}{2}\,\mathrm{m/s^2} = 3\sqrt{3}\,\mathrm{m/s^2}\,,$$
$$a_{1,y} = (6.00\,\mathrm{m/s^2})\sin(30^\circ) = 6.00 \cdot \frac{1}{2}\,\mathrm{m/s^2} = 3.00\,\mathrm{m/s^2}\,.$$

For the second thruster, we can write the angle as -60° because it is below the positive x-direction,

$$a_{2,x} = (2\sqrt{3}\,\mathrm{m/s^2})\cos(-60^\circ) = 2\sqrt{3} \cdot \frac{1}{2}\,\mathrm{m/s^2} = \sqrt{3}\,\mathrm{m/s^2}\,,$$
$$a_{2,y} = (2\sqrt{3}\,\mathrm{m/s^2})\sin(-60^\circ) = 2\sqrt{3} \cdot \left(-\frac{\sqrt{3}}{2}\right)\,\mathrm{m/s^2} = -3.00\,\mathrm{m/s^2}\,.$$

Now we add the vectors component-wise: Vector addition or subtraction should always be done by using the component form of the vectors. In most cases, adding the angles, or adding the magnitudes of the vectors, is incorrect.

Therefore, we add the two x-components together, and separately add the two y-components together,

$$a_x = a_{1,x} + a_{2,x} = (3\sqrt{3} + \sqrt{3})\,\mathrm{m/s^2} = 4\sqrt{3}\,\mathrm{m/s^2}\,,$$
$$a_y = a_{1,y} + a_{2,y} = (3.00 - 3.00)\,\mathrm{m/s^2} = 0\,\mathrm{m/s^2}\,.$$

Therefore, the total acceleration vector is

$$\vec{a} = (4\sqrt{3}\,\mathrm{m/s^2})\,\hat{x} + (0\,\mathrm{m/s^2})\,\hat{y} \approx (6.93\,\mathrm{m/s^2})\,\hat{x}\,.$$

Notice that the upward contribution from the first thruster exactly cancels the downward contribution from the second thruster. The horizontal contributions both point to the right, so they add together.

Part (b): Using the Pythagorean theorem, the magnitude of the total acceleration is

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2} = \sqrt{(4\sqrt{3})^2 + 0^2}\,\mathrm{m/s^2} = 4\sqrt{3}\,\mathrm{m/s^2} \approx 6.93\,\mathrm{m/s^2}\,.$$

Since $a_y = 0$ and $a_x > 0$, the total acceleration points in the positive x-direction, which makes an angle 0° with the positive x-axis. We can also check this using

$$\theta = \arctan\left(\frac{a_y}{a_x}\right) = \arctan\left(\frac{0}{4\sqrt{3}}\right) = 0^\circ\,.$$

Remember that we add the components first and then find the magnitude of the result. Simply adding the two given magnitudes would not give the magnitude of the total acceleration, because the two vectors point in different directions.

Problem 3 (continued)

Part (c): The second thruster now provides an acceleration with magnitude

$$|\vec{a}_2\,'| = \frac{1}{2}|\vec{a}_2| = \frac{1}{2}(2\sqrt{3}\,\mathrm{m/s^2}) = \sqrt{3}\,\mathrm{m/s^2}\,.$$

Its direction is unchanged, so its angle is still $-60°$. Using the same component formulas,

$$a'_{2,x} = (\sqrt{3}\,\mathrm{m/s^2})\cos(-60°) = \frac{\sqrt{3}}{2}\,\mathrm{m/s^2}\,,$$
$$a'_{2,y} = (\sqrt{3}\,\mathrm{m/s^2})\sin(-60°) = -\frac{3}{2}\,\mathrm{m/s^2}\,.$$

Both components of $\vec{a}_2$ are reduced by half. The components of $\vec{a}_1$ stay the same, so the new total acceleration has components

$$a'_x = a_{1,x} + a'_{2,x} = \left(3\sqrt{3} + \frac{\sqrt{3}}{2}\right)\,\mathrm{m/s^2} = \frac{7\sqrt{3}}{2}\,\mathrm{m/s^2}\,,$$
$$a'_y = a_{1,y} + a'_{2,y} = \left(3 - \frac{3}{2}\right)\,\mathrm{m/s^2} = \frac{3}{2}\,\mathrm{m/s^2}\,.$$

Thus the new total acceleration vector is

$$\vec{a}\,' = \left(\frac{7\sqrt{3}}{2}\,\mathrm{m/s^2}\right)\hat{x} + \left(\frac{3}{2}\,\mathrm{m/s^2}\right)\hat{y} \approx (6.06\,\mathrm{m/s^2})\,\hat{x} + (1.50\,\mathrm{m/s^2})\,\hat{y}\,.$$

Notice that we only reduced the second thruster's contribution by half. We should not divide the original *total* acceleration by two.

Part (d): Again using the Pythagorean theorem, we find

$$|\vec{a}\,'| = \sqrt{(a'_x)^2 + (a'_y)^2} = \sqrt{\left(\frac{7\sqrt{3}}{2}\right)^2 + \left(\frac{3}{2}\right)^2}\,\mathrm{m/s^2} = \sqrt{39}\,\mathrm{m/s^2} \approx 6.24\,\mathrm{m/s^2}\,.$$

Both components are positive, so the vector points up and to the right. Its angle above the positive x-direction is

$$\theta' = \arctan\left(\frac{a'_y}{a'_x}\right) = \arctan\left(\frac{3/2}{7\sqrt{3}/2}\right) = 13.9°\,.$$

The total acceleration no longer points purely in the positive x-direction. The second thruster now provides a smaller downward contribution, so it cannot cancel the upward contribution from the first thruster. The remaining y-component is positive, which agrees with the direction we just found. Make sure your calculator is in degree mode when computing this angle.